Who Won and Who Lost from the Tariff Shock?

AI-enhanced project brief – Option A

Work in pairs • Choose one option • 100 points • Deadline and submission link: see Canvas

Names _____ **Section** _____

Your decision question. Advise a non-partisan Canada–U.S. trade adjustment unit. Using protectionist measures implemented or in force from January 1, 2025 through October 31, 2026, rank the three sectors most plausibly hurt and the three strongest *protected or adapted candidates* across the two countries. Separate model predictions, measured exposure, and observed outcomes. A rigorous result may find no clear net winner. Recommend one targeted adjustment response and state what evidence would change the ranking.

1. Context and goal

The bilateral policy shock includes U.S. sectoral tariffs and Canadian counter-tariffs, with exemptions, remissions, policy changes, and supply-chain links. A tariff can protect a domestic producer while raising costs for downstream firms and consumers; retaliation can create a new set of exporters and importers at risk. "Protected" therefore does not automatically mean "better off."

Heckscher–Ohlin and Stolper–Samuelson give a long-run factor-income baseline: trade policy changes goods prices and therefore the real returns to factors used intensively in those goods. The specific-factors model is often the sharper short-run guide because plants, specialized capital, supplier networks, and places do not move immediately. Your task is to use both models without asking either to do more than it can.

This is an evidence-based incidence exercise, not a political position. Do not name or evaluate politicians. Freeze the policy register at October 31, 2026 so another researcher can reproduce your result even if policy later changes.

2. Required output – one package per pair

1. **One PDF** containing an exactly four-page narrative memo, followed by a references/data appendix of at most one page and a reproducibility/AI-use appendix of at most one page. Use 11-point type, ordinary margins, and no cover page.
2. **One research ZIP** containing a README, unchanged raw data, processed data, code, generated figures, a source audit, verification report, AI log, and partner contribution statement.
3. **Two principal exhibits** in the memo: one establishes incidence or exposure and one establishes outcomes or distribution. A compact decision table may supplement them if the memo remains readable.

One partner submits for the pair.
Use `LastName-LastName-memo.pdf` and `LastName-LastName-research.zip`. The appendices verify the analysis; they are not extra pages for arguments omitted from the four-page memo.

The Required Four-Page Memo

A short decision document, not a slide deck or literature review

3. Each page has one job

Page	Job	What the reader must learn
1	Finding and scope	Name the decision-maker, population or sectors, period, baseline, and comparison. State the answer and confidence in the first substantive paragraph.
2	Economic mechanism	Map the shock through the relevant model to named factors, sectors, firms, workers, or households. Explain the countervailing force.
3	Evidence and ranking	Present the two principal exhibits. Establish magnitudes and a defensible ranking; distinguish exposure, association, and causal evidence.
4	Decision and limits	Address the strongest contrary evidence, distribution, uncertainty, sensitivity, policy implication, next verification step, and reversal condition.

Answer template. "For [defined groups] over [period], the evidence most strongly identifies [finding]. I have [confidence level] because [reason]. I recommend [action], and would revise it if [observable evidence]."

4. Evidence and writing standard

- Put the answer first and write in complete paragraphs. Use claim-based headings rather than "Background" or a generic list of pros and cons.
- Give every exhibit a claim-based title, labelled axes and units, geography and dates, source, and note explaining transformations.
- Recreate figures from downloaded data. Do not paste website screenshots.
- Label each important claim as an observed fact, model prediction, association, causal estimate, scenario, or recommendation.
- Interpret each exhibit in prose and state its limitation. End with a decision and a concrete next check, not only "more research is needed."

Ottawa-ready bar. A decision-maker could act tomorrow because the memo states the magnitude, affected group, mechanism, uncertainty, policy implication, and the evidence that would change the recommendation.

Required Incidence Design

A winner–loser ranking needs a declared universe, rule, and time horizon

5. Build the ranking in four stages

1. **Freeze the policy register.** Record country, direction of trade, HS item, rate change, effective date, exemption or remission, and whether the measure was active on October 31, 2026.
2. **Choose a candidate universe.** Use 8–12 comparable sectors: include at least three Canadian and three U.S. directly exposed sectors, plus at least two downstream users of tariffed inputs. Declare exclusions first.
3. **Measure exposure.** Use pre-policy trade weights. A useful mechanical index is

$$E_s = \frac{\sum_{h \in s} M_{h,0}\, \Delta\tau_h}{Y_{s,0}},$$

   where $M_{h,0}$ is baseline bilateral imports and $Y_{s,0}$ is baseline sector shipments, value added, or payroll. This is exposure, not a causal effect.
4. **Measure outcomes.** Pre-register one main outcome—trade quantity or value, real shipments, employment, or real earnings—and one labour or price cross-check. Rank using a common percentage or sector-size denominator, not raw dollars alone.

6. Map the model to people and sectors

For every top-ranked sector, identify: protected domestic producers; foreign exporters; specific workers or owners; mobile factors; downstream input users; consumers; and the provinces, states, or industrial regions most exposed.

Required sensitivity. Recompute the ranking using both (1) a different reasonable denominator or pre-policy window and (2) a different outcome window. Report which ranks are stable. If your "winner" has positive protection but negative output or employment, explain rather than hiding the contradiction.

Language discipline. "High exposure," "rose after the tariff," and "caused by the tariff" are three different claims. A before–after line is not causal by itself; use a comparison sector, pre-trend, or published causal study when making a causal statement. Measures that began late in 2026 may be ranked on exposure, but do not claim an outcome with only a few post-policy weeks.

Evidence Plan

Reconcile tariff lines, trade flows, sector outcomes, and theory

7. Required exhibits and open-data starter pack

- **Exhibit 1 – exposure:** a Canada/U.S. sector chart showing the active policy rate, baseline trade value, exposure denominator, and resulting comparable exposure measure.
- **Exhibit 2 – outcomes:** event-time or indexed trends for the leading winners and losers, including a pre-policy period and at least one comparison sector. Mark policy dates and distinguish values from quantities.
- **Decision table:** top three losers and winners, country, mechanism, affected factor or group, observed evidence, confidence, and strongest alternative explanation.

Purpose	Official or research source
Canadian policy register	Finance Canada, complete counter-tariff list and steel and aluminum measures
U.S. policy and tariff lines	U.S. Harmonized Tariff Schedule; USITC Tariff Database; official proclamations, including the steel and aluminum action
Canadian trade by product	Statistics Canada CIMT Web Application, monthly customs-basis HS data
U.S. trade by product	USITC DataWeb, official Census merchandise trade statistics
Canadian outcomes	Statistics Canada Tables 14-10-0221-01 (employment/earnings), 16-10-0048-01 (manufacturing sales), and 18-10-0267-01 (producer prices)
U.S. outcomes	BLS Current Employment Statistics and Census Manufacturers' Shipments, Inventories, and Orders
External Canadian benchmarks	Bank of Canada, sector assessment, April 2026, and tariff assumptions, October 2026

Check import versus export direction; active versus superseded rates; exemptions, remissions, and CUSMA treatment; HS-year and HS–NAICS concordances; fixed pre-policy weights; nominal versus real values; seasonal adjustment; revisions; and at least three tariff lines against the official legal or tariff source.

Codex Research Workflow

Use AI to accelerate auditable inquiry, not to outsource judgment

8. Use a dedicated project folder

```
project/
  README.md    sources.md    AI_LOG.md
  data/raw/    data/processed/    code/
  figures/    memo/    verification_report.md
```

Preserve downloaded files unchanged in `data/raw`. Generated files belong in `data/processed`. Never put passwords, tokens, private student data, or confidential information in a prompt.

9. A one-week research-assistant sequence

Day	Objective	Required practice
1	Work backward	Freeze the scope; define groups, baseline, comparison, mechanism, ranking rule, needed evidence, and possible refutations.
2	Acquire and audit	Download primary files; record definitions, units, classifications, geography, dates, access dates, and revision status.
3	Build measures	Write reproducible code; test joins, missingness, duplicates, totals, units, and identities. Manually verify three observations.
4	Build exhibits	Create the ranking and two exhibits; test an alternative denominator, window, or pass-through assumption.
5	Draft	Write the finding first, then mechanism, evidence, contrary case, decision, and reversal condition.
6	Red-team	Search for direction errors, unsupported causal language, fragile ranks, and evidence inconsistent with the preferred conclusion.
7	Reproduce	Run from raw data in a clean folder; regenerate every exhibit; audit the PDF, links, sources, AI log, and contributions.

10. Useful prompts

Plan: "Do not answer yet. Turn the assigned question into groups, mechanisms, measures, competing explanations, a ranking rule, and observations that could overturn each claim."
Data: "Inspect the documentation before coding. Build a data dictionary from actual columns. Flag classification joins, incompatible units or dates, revisions, missing values, and policy-direction errors. Invent nothing."
Audit: "Classify each material claim as fact, model prediction, association, causal estimate, scenario, or recommendation. Flag unsupported claims, denominator errors, fragile ranks, and exhibits that do not support the prose. Do not rewrite the memo yet."

AI output is not a source. You remain responsible for every number, citation, inference, and sentence. Record important retained output, errors, corrections, and each partner's contribution in the AI-use appendix.

Rubric and Submission Gate

The bar is a concise document that another researcher can audit

11. Assessment rubric – 100 points

Dimension	Weight	What level 5 demonstrates
Exact question and answer-first finding	15	Direct answer with defined groups, dates, baseline, comparison, confidence, and conditions.
Economic mechanism and incidence	20	Correct model mapped to named factors, sectors, regions, workers, consumers, or households.
Measurement and empirical design	20	Valid open-data measures; consistent units and classifications; descriptive and causal claims clearly separated.
Evidence, rankings, and exhibits	20	Two reproducible exhibits establish magnitudes and a defensible ranking that survives a meaningful sensitivity.
Judgment and Ottawa-ready usefulness	15	Contrary evidence, distribution, uncertainty, action, next check, and reversal condition are decision-relevant.
Professional memo, reproducibility, and AI transparency	10	Four readable pages, complete source trail, clean rerunnable code, preserved raw data, AI corrections, and contributions.

12. Levels and hard gates

Each dimension receives a level: **5 decision-ready and independently auditable**; **4 strong**; **3 correct and useful, but not yet decision-ready**; **2 partial or materially flawed**; **1 minimal**; or **0 missing or unsupported**. Row points equal weight × level/5.

No verified open data or reproducible code: overall maximum 50. An adjacent essay that does not answer the question: first row 0 and overall maximum 50. Theory named but not mapped to incidence: mechanism maximum level 2. Exposure or correlation presented as causal: design maximum level 2. No meaningful sensitivity: evidence maximum level 3. Fabricated or untraceable evidence: evidence and reproducibility receive 0. No distributional analysis or serious contrary case: judgment maximum level 2.

13. Verification and final upload

Every number traces to a dataset, variable, and raw observation; classifications, dates, currencies, prices, units, and denominators are consistent or disclosed; three observations and one official aggregate were checked by hand; raw data remain unchanged; exclusions and revisions are documented; all exhibits regenerate from a clean run; the required sensitivity is reported; and both partners can explain the memo, code, figures, and AI corrections.

Submit one memo PDF and one research ZIP through Canvas, then reopen both. Canvas is the authoritative deadline and submission record.

Final test. Could a skeptical policy reader reproduce the exhibits, understand exactly who gains and loses, see the best case against the finding, and know what evidence would change the decision?